B.TECH ENGINEERING MATHEMATICS · SEMESTER I

Differential Equations

Unit I – First Order & First Degree · Complete Formula Sheet

SUBJECT	LEVEL	PURPOSE
Engineering Mathematics	**B.Tech · Year 1, Sem 1**	**Last-Day Exam Revision**

CONTENTS

FORMULA SHEET · NOT FOR COMMERCIAL USE · EXAM REVISION ONLY

📘 Basic Concepts & Definitions

SECTION 01

CORE DEFINITIONS

Differential Equation (DE): An equation involving an independent variable, a dependent variable, and derivatives of the dependent variable w.r.t. the independent variable.

Order: The order of the *highest* derivative present in the equation.

Degree: The power (exponent) of the highest-order derivative, after clearing fractions and radicals.

General Solution: Solution containing as many arbitrary constants as the order of the DE.

Particular Solution: Obtained by giving specific values to the arbitrary constants using initial/boundary conditions.

Formation of DE: Differentiate the given relation as many times as the number of arbitrary constants, then eliminate the constants.

QUICK IDENTIFICATION

ORDER

= Highest derivative in DE

e.g. d^3y/dx^3 → order = 3

DEGREE

= Power of highest-order derivative

e.g. $(dy/dx)^2$ → degree = 2

✂️ Variables Separable

SECTION 02

STANDARD FORM

FORM

$$f(x)\,dx + g(y)\,dy = 0$$

Can also appear as: $dy/dx = f(x) \cdot g(y)$

METHOD (STEPS)

1. Separate variables: $f(x)\,dx = -g(y)\,dy$
2. Integrate both sides: $\int f(x)\,dx = \int -g(y)\,dy + C$
3. Simplify and write solution.

SOLUTION TEMPLATE

$$\int f(x)\,dx + \int g(y)\,dy = C$$

🔃 Homogeneous Equations

SECTION 03

STANDARD FORM

FORM

$dy/dx = f(y/x)$ or $Mdx + Ndy = 0$ where M, N are homogeneous of same degree

METHOD (STEPS)

1. Put $y = vx \rightarrow dy/dx = v + x\,dv/dx$
2. Substitute → equation becomes separable in v and x
3. Separate & integrate: $\int dv / [f(v)-v] = \int dx/x + C$
4. Back-substitute $v = y/x$

KEY SUBSTITUTION

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y = vx → dy/dx = v + x(dv/dx)
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📐 Linear Differential Equations

SECTION 04

STANDARD FORM (IN Y)

STANDARD FORM

$$dy/dx + P(x) \cdot y = Q(x)$$

P and Q are functions of x only

INTEGRATING FACTOR & SOLUTION

INTEGRATING FACTOR (I.F.)

$$\text{I.F.} = e^{\int P\,dx}$$

GENERAL SOLUTION

$$y \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.})\,dx + C$$

STANDARD FORM (IN X)

ALTERNATIVE FORM (X AS DEPENDENT)

$$dx/dy + P(y) \cdot x = Q(y)$$

I.F. = $e^{\int P\,dy}$ · Solution: $x \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.})\,dy + C$

STEP-BY-STEP

1. Write in standard form: $dy/dx + Py = Q$
2. Find $\text{I.F.} = e^{\int P\,dx}$
3. Multiply both sides by I.F.
4. L.H.S. becomes $d/dx[y \cdot \text{I.F.}]$
5. Integrate R.H.S.: $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\,dx + C$

🔁 Bernoulli's Equation

SECTION 05

STANDARD FORM

BERNOULLI FORM

$$dy/dx + P(x) \cdot y = Q(x) \cdot y^n$$

When $n = 0$ → Linear DE; when $n = 1$ → Variables separable

REDUCTION STEPS

1. Divide throughout by y^n : $y^{-n} \cdot dy/dx + P \cdot y^{(1-n)} = Q$
2. Substitute: $v = y^{(1-n)}$
3. Then: $dv/dx = (1-n) \cdot y^{(-n)} \cdot dy/dx$
4. Equation becomes: $dv/dx + (1-n)P \cdot v = (1-n)Q$ → **Linear in v!**
5. Solve as Linear DE, then back-substitute $v = y^{(1-n)}$

KEY SUBSTITUTION

$$v = y^{(1-n)} \rightarrow dv/dx = (1-n) \cdot y^{(-n)} \cdot dy/dx$$

This converts Bernoulli → Linear DE

LINEAR VS BERNOULLI QUICK COMPARISON

FEATURE	LINEAR DE	BERNOULLI'S DE
Form	$dy/dx + Py = Q$	$dy/dx + Py = Qy^n$
R.H.S.	Function of x only	Has y^n factor ($n \neq 0,1$)
Substitution	None needed	$v = y^{(1-n)}$
I.F.	$e^{\int P\,dx}$	After reducing to linear

FEATURE	LINEAR DE	BERNOULLI'S DE
Solution	$y \cdot I.F. = \int Q \cdot I.F.\, dx + C$	$v \cdot I.F. = \int (1-n) Q \cdot I.F.\, dx + C$

Exact Differential Equations

SECTION 06

STANDARD FORM & CONDITION

STANDARD FORM

$M(x,y)\ dx + N(x,y)\ dy = 0$

EXACTNESS CONDITION

$\partial M/\partial y = \partial N/\partial x$

Must be checked first!

SOLUTION METHOD

1. Verify: $\partial M/\partial y = \partial N/\partial x$
2. Integrate M w.r.t. x (treat y as constant): $F = \int M\ dx$
3. Differentiate F w.r.t. y: $\partial F/\partial y$
4. Set $\partial F/\partial y = N$ → find the function $g(y)$
5. Write complete solution: $\int M\ dx + \int(\text{terms in N not in } \partial/\partial y \int M\ dx)\ dy = C$

SOLUTION FORMULA

$\int M\ dx + \int(\text{terms in N free of x})\ dy = C$

(treating y as constant in the first integral)

Equations Reducible to Exact Form

SECTION 07

When $Mdx + Ndy = 0$ is NOT exact, multiply by an Integrating Factor (μ) to make it exact.

CASE 1 – I.F. AS FUNCTION OF X ONLY

CONDITION

$(\partial M/\partial y - \partial N/\partial x)\ /\ N = f(x)$ [function of x only]

INTEGRATING FACTOR

$\mu(x) = e^{\wedge}(\int f(x)\ dx)$

CASE 2 – I.F. AS FUNCTION OF Y ONLY

CONDITION

$(\partial N/\partial x - \partial M/\partial y)\ /\ M = g(y)$ [function of y only]

INTEGRATING FACTOR

$\mu(y) = e^{\wedge}(\int g(y)\ dy)$

SUMMARY TABLE – INTEGRATING FACTOR CASES

CASE	EXPRESSION	DEPENDS ON	I.F. FORMULA
Case 1	$(\partial M/\partial y - \partial N/\partial x)\ /\ N$	x only → f(x)	$e^{\wedge}(\int f(x)\ dx)$
Case 2	$(\partial N/\partial x - \partial M/\partial y)\ /\ M$	y only → g(y)	$e^{\wedge}(\int g(y)\ dy)$
Special	$M = yf(xy),\ N = xg(xy)$	xy combination	$1\ /\ (xM - yN)$

EXACT VS NON-EXACT QUICK COMPARISON

PROPERTY	EXACT DE	NON-EXACT DE
Condition	$\partial M/\partial y = \partial N/\partial x$	$\partial M/\partial y \neq \partial N/\partial x$
Action	Solve directly	Find I.F. first, then solve

PROPERTY	EXACT DE	NON-EXACT DE
Solution exists	$F(x,y) = C$	After multiplying by μ

Applications of First-Order DEs

SECTION 08

NEWTON'S LAW OF COOLING

DE FORM

$$dT/dt = -k(T - T_s)$$

$k > 0$, T_s = surrounding temperature

GENERAL SOLUTION

$$T(t) = T_s + (T_0 - T_s) \cdot e^{(-kt)}$$

T_0 = initial temperature at $t = 0$

KEY POINTS

- $T \to T_s$ as $t \to \infty$
- k found from extra condition
- Separation of variables method

INTEGRATED FORM (SEPARATION)

$$\ln|T - T_s| = -kt + C$$

LAW OF NATURAL GROWTH & DECAY

DE FORM

$$dy/dt = ky$$

$k > 0$: growth; $k < 0$: decay

SOLUTION

$$y(t) = y_0 \cdot e^{(kt)}$$

y_0 = initial value at $t = 0$

HALF-LIFE (DECAY)

$$t_{1/2} = \ln 2 / |k|$$

When $y = y_0/2$

ELECTRICAL CIRCUITS (RL SERIES)

DE (RL CIRCUIT)

$$L \cdot dI/dt + R \cdot I = E(t)$$

L = inductance, R = resistance, E = EMF

STANDARD LINEAR FORM

$$dI/dt + (R/L) \cdot I = E(t)/L$$

$P = R/L$, $Q = E(t)/L$

I.F. AND SOLUTION

$$\text{I.F.} = e^{(Rt/L)}$$

$I \cdot e^{(Rt/L)} = \int (E/L) \cdot e^{(Rt/L)}\, dt + C$

STEADY STATE (DC, $E = E_0$)

$$I(t) = (E_0/R) + Ce^{(-Rt/L)}$$

As $t \to \infty$: $I \to E_0/R$ (steady state)

RC CIRCUIT (CHARGE)

DE FOR CHARGE Q ON CAPACITOR

$$R \cdot dQ/dt + Q/C = E(t) \to dQ/dt + (1/RC) \cdot Q = E(t)/R$$

$I = dQ/dt$; C = capacitance; Charge: $Q(t) = CE + (Q_0 - CE) \cdot e^{(-t/RC)}$

APPLICATION FORMULAS – SUMMARY TABLE

APPLICATION	DE FORM	SOLUTION	KEY CONSTANT
Newton's Cooling	$dT/dt = -k(T-T_s)$	$T = T_s + (T_0-T_s)e^{(-kt)}$	k (cooling rate)
Growth	$dy/dt = ky,\ k>0$	$y = y_0e^{(kt)}$	k (growth rate)
Decay	$dy/dt = -ky,\ k>0$	$y = y_0e^{(-kt)}$	k (decay rate)
RL Circuit	$L \cdot dI/dt + RI = E$	$I = E/R + Ce^{(-Rt/L)}$	$\tau = L/R$ (time constant)
RC Circuit	$R \cdot dQ/dt + Q/C = E$	$Q = CE + (Q_0-CE)e^{(-t/RC)}$	$\tau = RC$ (time constant)

Master Summary & Comparison Tables

SECTION 09

TYPES OF FIRST ORDER FIRST DEGREE DES – MASTER TABLE

#	TYPE	IDENTIFICATION	KEY SUBSTITUTION	I.F. / METHOD	SOLUTION FORM
1	**Variables Separable**	f(x)dx + g(y)dy = 0	None	Direct integration	∫f(x)dx + ∫g(y)dy = C
2	**Homogeneous**	M, N same degree in x,y	y = vx	Separable after sub.	In terms of v = y/x
3	**Linear (y)**	dy/dx + Py = Q	None	e^(∫P dx)	y·I.F. = ∫Q·I.F. dx + C
4	**Linear (x)**	dx/dy + Px = Q	None	e^(∫P dy)	x·I.F. = ∫Q·I.F. dy + C
5	**Bernoulli**	dy/dx + Py = Qy^n	v = y^(1–n)	Reduce → Linear	Solve linear in v
6	**Exact**	∂M/∂y = ∂N/∂x	None	Direct	∫M dx + ∫(N–∂/∂y∫M dx) dy = C
7	**Non-Exact (I.F. of x)**	(∂M/∂y–∂N/∂x)/N = f(x)	None	μ = e^(∫f dx)	Multiply, then exact
8	**Non-Exact (I.F. of y)**	(∂N/∂x–∂M/∂y)/M = g(y)	None	μ = e^(∫g dy)	Multiply, then exact

DECISION FLOWCHART (TEXT VERSION)

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Given: dy/dx = f(x,y) or Mdx + Ndy = 0

Step 1: Can you write f(x)dx + g(y)dy = 0? → Variables Separable
Step 2: Is M, N homogeneous of same degree? → Homogeneous (put y=vx)
Step 3: Is form dy/dx + Py = Q (no yⁿ)? → Linear DE
Step 4: Is form dy/dx + Py = Qyⁿ? → Bernoulli (put v = y^(1−n))
Step 5: Check ∂M/∂y = ∂N/∂x → Exact; else find I.F. → Reducible to Exact
```

ALL KEY FORMULAS AT A GLANCE

LINEAR DE – I.F.
I.F. = e^(∫P dx)

EXACT – CONDITION
∂M/∂y = ∂N/∂x

LINEAR DE – SOLUTION
y · I.F. = ∫Q · I.F. dx + C

I.F. CASE 1 (F OF X)
μ = e^(∫[(∂M/∂y–∂N/∂x)/N] dx)

BERNOULLI – SUBSTITUTION
v = y^(1–n)

I.F. CASE 2 (F OF Y)
μ = e^(∫[(∂N/∂x–∂M/∂y)/M] dy)

BERNOULLI – REDUCED FORM
dv/dx + (1–n)Pv = (1–n)Q

HOMOGENEOUS – SUB
y = vx, dy/dx = v + x·dv/dx

Exam Flashcards

30 CARDS

Q01 What is a Differential Equation?

A Equation with dependent var, independent var, and its derivatives

Q02 What is the ORDER of a DE?

A Order = Highest derivative present in the equation

Q03 What is the DEGREE of a DE?

A Power of the highest-order derivative (after clearing radicals)

Q04 General Solution vs Particular Solution?

A General: has arbitrary constants. Particular: constants found from I.C.

Q05 Variables Separable form?

A $f(x)dx + g(y)dy = 0 \rightarrow \int f(x)dx + \int g(y)dy = C$

Q06 Homogeneous DE substitution?

A $y = vx \rightarrow dy/dx = v + x\cdot(dv/dx)$

Q07 Linear DE standard form?

A $dy/dx + P(x)\cdot y = Q(x)$

Q08 Integrating Factor for Linear DE?

A I.F. = $e^{\wedge}(\int P\ dx)$

Q09 Solution of Linear DE?

A $y \cdot$ I.F. = $\int Q \cdot$ I.F. $dx + C$

Q10 Bernoulli's Equation form?

A $dy/dx + P(x)\cdot y = Q(x)\cdot y^n$

Q11 Bernoulli substitution?

A $v = y^{\wedge}(1-n)$ → reduces to Linear DE

Q12 After Bernoulli substitution, dv/dx =?

A $dv/dx = (1-n) \cdot y^{\wedge}(-n) \cdot dy/dx$

Q13 Exact DE standard form?

A $M(x,y)dx + N(x,y)dy = 0$

Q14 Exactness condition?

A $\partial M/\partial y = \partial N/\partial x$

Q15 Solution of Exact DE?

A $\int M\ dx + \int$(terms in N free of x) $dy = C$

Q16 I.F. when $(\partial M/\partial y - \partial N/\partial x)/N = f(x)$?

A $\mu = e^{\wedge}(\int f(x)\ dx)$

Q17 I.F. when $(\partial N/\partial x - \partial M/\partial y)/M = g(y)$?

A $\mu = e^{\wedge}(\int g(y)\ dy)$

Q18 Newton's Law of Cooling DE?

A $dT/dt = -k(T - T_s)$, $k > 0$

Q19 Newton's Cooling solution?

A $T(t) = T_s + (T_0 - T_s)\cdot e^{\wedge}(-kt)$

Q20 Natural Growth/Decay DE?

A $dy/dt = ky \rightarrow y = y_0\cdot e^{\wedge}(kt)$

Q21 Growth vs Decay – how to tell?

A $k > 0$ → growth; $k < 0$ → decay

Q22 Half-life formula?

A $t_{1/2} = \ln(2) / |k|$

Q23 RL Circuit DE?

A $L\cdot dI/dt + R\cdot I = E(t)$

Q24 Steady-state current in RL circuit?

A $I\_ss = E_0/R$ (as $t \rightarrow \infty$)

Q25 If n=0 in Bernoulli, what type is it?

A Linear DE ($y^n = y^0 = 1$, no y term on RHS)

Q26 If n=1 in Bernoulli, what type is it?

A Variables Separable ($dy/dx + Py = Qy$)

Q27 How to identify Homogeneous DE?

A $M(tx,ty) = t^n M(x,y)$ and $N(tx,ty) = t^n N(x,y)$ same degree

Q28 Bernoulli reduced linear form?

A $dv/dx + (1-n)P\cdot v = (1-n)Q$

Q29 L.H.S. after multiplying by I.F. in Linear DE?

A d/dx [$y \cdot$ I.F.] (perfect derivative)

Q30 RC Circuit charge equation?

A $Q(t) = CE + (Q_0 - CE)\cdot e^{\wedge}(-t/RC)$

⚠️ Exam Tips & Common Mistakes

SECTION 11

CRITICAL MISTAKES TO AVOID

- ❌ **Confusing Linear and Bernoulli:** If R.H.S. has y^n ($n \neq 0,1$), it's Bernoulli — NOT linear. Always check for y^n on R.H.S. before deciding the type.
- ❌ **Forgetting I.F. Formula:** For linear DE, I.F. = e^(∫P dx). Don't compute e^(P·x) — you must integrate P first, then exponentiate!
- ❌ **Skipping the Exactness Check:** Always verify $\partial M/\partial y = \partial N/\partial x$ before attempting to solve as exact. Attempting without checking wastes time.
- ❌ **Wrong Bernoulli Substitution:** v = y^(1–n), NOT y^(n–1). The exponent is (1 MINUS n). Get the sign right every time.
- ❌ **Sign Error in Newton's Cooling:** The DE is dT/dt = **–k**($T–T_s$), negative sign because the body cools (temperature decreases). Don't drop the negative sign!
- ❌ **Growth vs Decay Confusion:** Growth → +k (y increases). Decay → –k (y decreases). In problems: k is always given as positive; the sign of the DE determines growth/decay.
- ❌ **Forgetting +C in Integration:** Every indefinite integral produces a constant C. In final solutions, always include + C (or evaluate it from initial conditions).
- ❌ **Wrong I.F. Case Selection:** For Case 1, divide by N (not M). For Case 2, divide by M (not N). The expressions are: $(\partial M/\partial y - \partial N/\partial x)/N$ for Case 1 and $(\partial N/\partial x - \partial M/\partial y)/M$ for Case 2.
- ❌ **Incorrect Partial Differentiation:** When finding $\partial M/\partial y$, differentiate M with respect to y, treating x as constant. And for $\partial N/\partial x$, differentiate N w.r.t. x, treating y as constant. Don't swap these!
- ❌ **Not Back-Substituting:** After solving Bernoulli in terms of v, always replace v = y^(1–n) to get the solution in terms of x and y.

✅ PRO STRATEGIES

- ✔ Always write DE in standard form first before identifying type.
- ✔ In exact DE solution, integrate M w.r.t. x first, then handle remaining terms from N.
- ✔ For circuit problems, always identify R, L, C and write standard linear form immediately.
- ✔ Write I.F. = e^(∫P dx) clearly, then simplify the exponent before writing e^(...).
- ✔ In applications, write the DE first, then solve — don't jump to the solution formula.
- ✔ Check final answer by substituting back into original DE when time permits.

Quick Reference Sheet

LAST-MINUTE REVISION

DEFINITIONS (ONE LINE EACH)

- **DE** Equation with variable + derivative
- **Order** Highest derivative order
- **Degree** Power of highest derivative
- **General Sol.** Contains arbitrary constants
- **Particular Sol.** Constants determined by I.C.
- **Exact DE** $\partial M/\partial y = \partial N/\partial x$
- **I.F.** Multiplier to make exact
- **Homogeneous** M, N of same degree

ALL KEY FORMULAS

VARIABLES SEPARABLE

$\int f(x)dx + \int g(y)dy = C$

HOMOGENEOUS SUB

$y = vx$

$dy/dx = v + x \cdot dv/dx$

LINEAR I.F.

$I.F. = e^{\int P\,dx}$

LINEAR SOLUTION

$y \cdot I.F. = \int Q \cdot I.F.\,dx + C$

BERNOULLI SUB

$v = y^{(1-n)}$

BERNOULLI REDUCED

$dv/dx+(1-n)Pv=(1-n)Q$

EXACT CONDITION

$\partial M/\partial y = \partial N/\partial x$

I.F. CASE 1

$\mu = e^{\int [(\partial M/\partial y-\partial N/\partial x)/N]dx}$

I.F. CASE 2

$\mu = e^{\int [(\partial N/\partial x-\partial M/\partial y)/M]dy}$

APPLICATION EQUATIONS

NEWTON'S COOLING

$T = T_s + (T_0-T_s)e^{(-kt)}$

GROWTH / DECAY

$y = y_0 \cdot e^{(kt)}$

RL CIRCUIT

$I = E/R + Ce^{(-Rt/L)}$

STANDARD FORMS SUMMARY

TYPE	STANDARD FORM	I.F. / SUB
Linear	$dy/dx + Py = Q$	$e^{\int P\,dx}$
Bernoulli	$dy/dx + Py = Qy^n$	$v = y^{(1-n)}$
Exact	$Mdx + Ndy = 0,\ \partial M/\partial y=\partial N/\partial x$	Direct
Homogeneous	$dy/dx = f(y/x)$	$y = vx$
Sep. Variables	$f(x)dx + g(y)dy = 0$	Direct $\int$